

AIM:

Upload and analyse the given dataset and perform data preprocessing and visualisation.

PYTHON CODE:

```
import pandas as pd
import numpy as np

df = pd.read_excel("Hotel_Dataset.xlsx")

print("Original Dataset:")
print(df.head())

print("\nDataset Information:")
df.info()

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Records:")
print(df.duplicated().sum())

df.drop_duplicates(inplace=True)

numeric_columns = df.select_dtypes(include=np.number).columns

for column in numeric_columns:
    df[column] = df[column].fillna(df[column].median())

categorical_columns = df.select_dtypes(include="object").columns

for column in categorical_columns:
    if df[column].isnull().any():
        df[column] = df[column].fillna(df[column].mode()[0])

for column in numeric_columns:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    df[column] = df[column].clip(lower_limit, upper_limit)

print("\nMissing Values After Preprocessing:")
print(df.isnull().sum())
```

```
print("\nDuplicate Records After Preprocessing:")
print(df.duplicated().sum())

print("\nPreprocessed Dataset:")
print(df.head())

df.to_excel("Preprocessed_Hotel_Dataset.xlsx", index=False)

print("\nData preprocessing completed successfully.")
```

OUTPUT:

Original Dataset:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
Age_Group.1								
0	1	20-25	4	Ibis	veg 1300 2	40000	20-25	
1	2	30-35	5	LemonTree	Non-Veg 2000 3	59000	30-35	
2	3	25-30	6	RedFox	Veg 1322 2	30000	25-30	
3	4	20-25	-1	LemonTree	Veg 1234 2	120000	20-25	
4	5	35+	3	Ibis	Vegetarian 989 2	45000	35+	

Dataset Information:

<class 'pandas.DataFrame'>

RangeIndex: 11 entries, 0 to 10

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
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0	CustomerID	11 non-null	int64
1	Age_Group	11 non-null	str
2	Rating(1-5)	11 non-null	int64
3	Hotel	11 non-null	str
4	FoodPreference	11 non-null	str
5	Bill	11 non-null	int64
6	NoOfPax	11 non-null	int64
7	EstimatedSalary	11 non-null	int64
8	Age_Group.1	11 non-null	str

dtypes: int64(5), str(4)

memory usage: 924.0 bytes

Missing Values:

CustomerID	0
Age_Group	0
Rating(1-5)	0
Hotel	0
FoodPreference	0
Bill	0
NoOfPax	0

```
EstimatedSalary    0
Age_Group.1        0
dtype: int64
```

Duplicate Records:

1

/Users/yuva/Desktop/Code_Hub/FDS_Exp/Exp_3/exp.py:25: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to 'include' to select them, or to 'exclude' to remove them and silence this warning.

See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.

```
categorical_columns = df.select_dtypes(include="object").columns
```

Missing Values After Preprocessing:

```
CustomerID        0
Age_Group          0
Rating(1-5)       0
Hotel             0
FoodPreference    0
Bill              0
NoOfPax           0
EstimatedSalary   0
Age_Group.1       0
dtype: int64
```

Duplicate Records After Preprocessing:

0

Preprocessed Dataset:

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
Age_Group.1								
0	1	20-25	4	Ibis	veg	1300.0	2.0	40000.0
1	2	30-35	5	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3	25-30	6	RedFox	Veg	1322.0	2.0	30000.0
3	4	20-25	0	LemonTree	Veg	1234.0	2.0	120000.0
4	5	35+	3	Ibis	Vegetarian	989.0	2.0	45000.0

Data preprocessing completed successfully.

RESULT:

The given dataset is analysed and preprocessed.